

ECON-1020 Introductory Macroeconomics

Chapter 4 The Market Forces of Supply and Demand

Shutao Cao

2025-09-08

Trent University

Fall 2025

Outline

Competitive Market

Demand

Supply

Market Equilibrium

Equilibrium Analysis

Concluding remarks

Competitive Market

Demand

Supply

Market Equilibrium

Equilibrium Analysis

Concluding remarks

Nothing is less real than realism. Details are confusing. It is only by selection, by elimination, by emphasis, that we get at the real meaning of things.

—Georgia O’Keeffe (1887 - 1986)

- How do people make choices?
 - Preference, what and how much people want of a good
 - Budget constraint: how much money people allocate to the purchase, and prices of goods
- The outcome is the amount of a good that people want to buy and can afford to at each price level
- This choice-making calculation leads to the demand curve
- Similar decisions on supply are made by producers

Reading of the week

This week

- Textbook Chapter 4
- Textbook Chapter 4 Appendix

Next week

- Chapter 5

Competitive Market

A **market** is an arrangement in which buyers and sellers collect information and trade with each other a particular good or service

- Some markets are highly organized, e.g. stock market, commodity markets, where there is an auctioneer that sets price to match sellers and buyers
- Many markets are less organized, e.g. market for ice-cream in Peterborough, corporate bond market, the labor market

A **competitive market** is a market with many buyers and sellers of the same good or service, each single buyer or seller has negligible influence on the market price

A **perfectly competitive market** has the following properties:

- Homogeneous good, i.e., all goods are identical
- No single buyer or seller has any influence over the market price
- Complete information, i.e., everyone knows the market demand and supply
- Zero transaction cost

Demand

Demand

The **quantity demanded** of a good is the amount of the good that buyers are willing to and able to purchase at a given price

The **demand** for a good describes how the quantity demanded varies when the price of the good changes

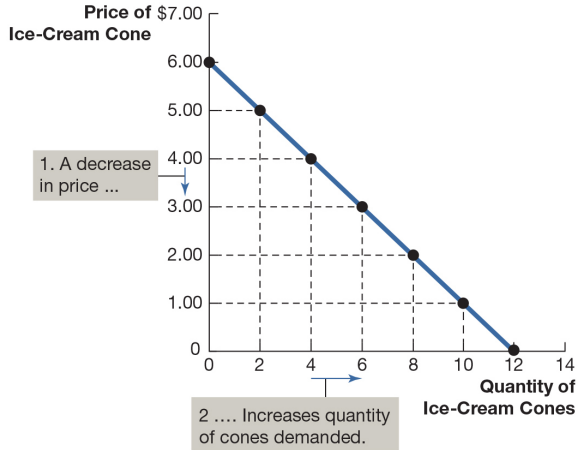
The **demand curve** is a graph of the relationship between the price of a good and the quantity demanded

The demand for a good reflects:

1. wants for the good (preference);
2. the ability to pay for a wanted amount of the good

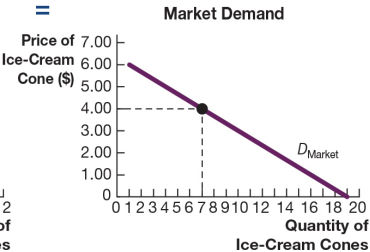
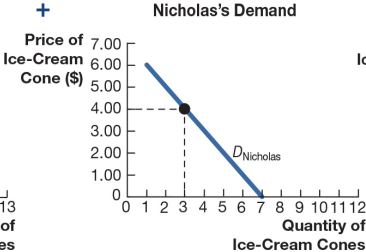
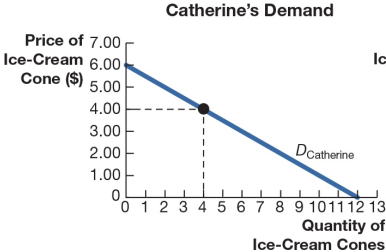
The **demand schedule** is a table showing how much of a good or service consumers buy at different prices, *holding constant everything else*

Price of Ice-Cream Cone	Catherine
\$0.00	12
1.00	10
2.00	8
3.00	6
4.00	4
5.00	2
6.00	0



Individual demand and market demand

Price of Ice-Cream Cone	Catherine		Nicholas		Market
\$0.00	12	+	7	=	19
1.00	10		6		16
2.00	8		5		13
3.00	6		4		10
4.00	4		3		7
5.00	2		2		4
6.00	0		1		1



The Law of Demand

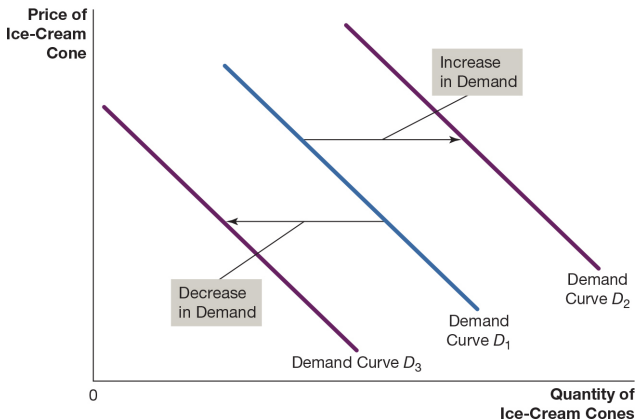
Other things remaining unchanged, when the price of a good rises, the quantity demanded of the good falls; when the good's price falls, the quantity demanded rises

- It is called a law because it is widely observed

Shifts in the demand curve

A rightward shift in the demand curve is an **increase in demand**, vice versa

A leftward shift in the demand curve is a **decrease in demand**, vice versa

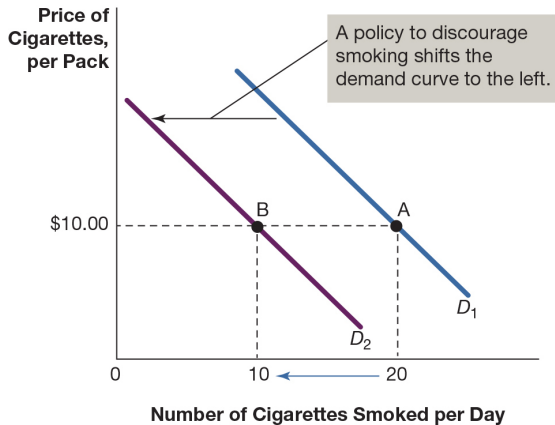


Shifts versus Movements

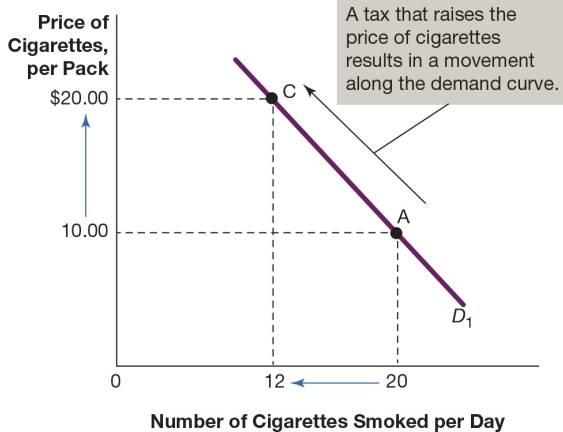
A shift of the demand curve is a change in quantity demanded given each price. The shift is caused by a change in "other things"

A movement along the demand curve is a change in quantity demanded when price changes, holding "other things" constant

(a) A Shift in the Demand Curve



(b) A Movement along the Demand Curve



What cause the demand curve to shift?

Shifts in the demand curve are caused by factors other than the good's price

These factors include changes in

- the prices of related goods
- income
- tastes
- expectations
- the number of consumers

Changes in the prices of related goods: substitutes

Two goods are **substitutes** if a rise in the price of one good leads to an increase in the demand for the other

- Goods serving a similar function, e.g., heating oil and natural gas; coffee and tea
- When the price of a good rises, the demand for its substitute increases, demand curve for the substitute shifts to the right
- When the price of a good falls, the demand for its substitute decreases, demand curve for the substitute shifts to the left

Changes in the prices of related goods: complements

Two goods are **complements** if a rise in the price of one good leads to a decrease in the demand for the other

- E.g., computer and software; cars and gasoline
- When the price of a good rises, the demand for its complement decreases, demand curve for the complement shifts to the left
- When the price of a good drops, the demand for its complement increases, demand curve for the complement shifts to the right

Changes in income

A good is a **normal good** if its demand increases when consumer income rises

- E.g., a high-end restaurant
- When income rises, demand curve of a normal good shifts to the right

A good is an **inferior good** if its demand declines when consumer income rises

- E.g., fast-food chains; public transportation
- When income rises, demand curve for an inferior good shifts to the left

Changes in tastes

People change tastes

The demand curve for a good shifts if tastes for that good change

- Change in consumption patterns forced by the pandemic may lead to change in preference

Changes in expectations

Anticipating a future drop in price leads to a decrease in demand today, shifting the demand curve today to the left

Changes in the number of consumers

When there are more consumers, market demand increases, shifting the market demand curve to the *right*

Spacer: examples in classroom

Supply

Supply

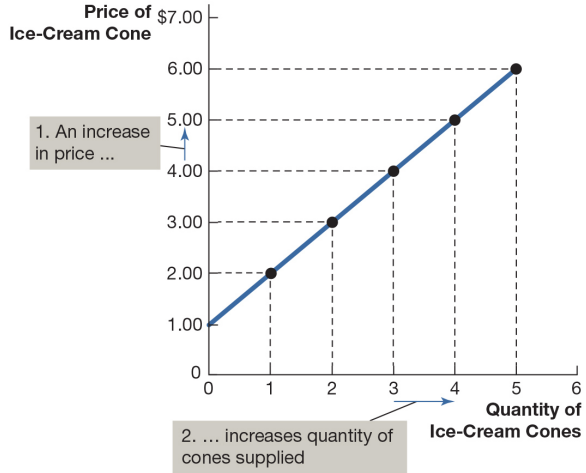
The **quantity supplied** of a good is the quantity that producers are willing to and able to sell at a given price

The **supply** of a good is the relationship between the quantity supplied and the price of the good, when all other influences on producers' sales remain constant

- The supply of a good can be represented by the **supply curve**

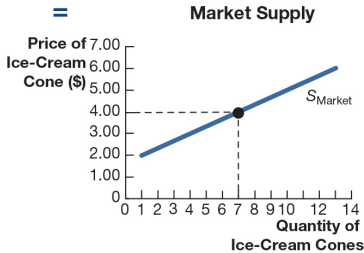
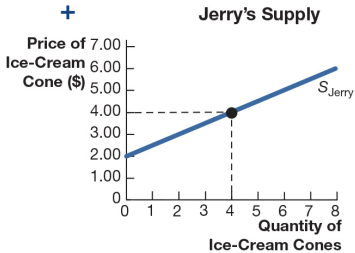
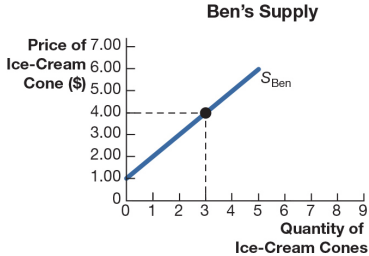
The supply schedule is a table showing quantity supplied and prices

Price of Ice-Cream Cone	Quantity of Cones Supplied
\$0.00	0
1.00	0
2.00	1
3.00	2
4.00	3
5.00	4
6.00	5



Market supply is the horizontal sum of individuals supplies

Price of Ice-Cream Cone	Ben		Jerry		Market
\$0.00	0	+	0	=	0
1.00	0		0		0
2.00	1		0		1
3.00	2		2		4
4.00	3		4		7
5.00	4		6		10
6.00	5		8		13



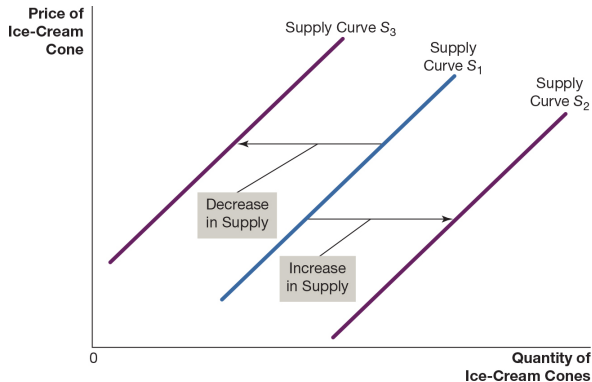
The Law of Supply

Other things remaining constant, the quantity supplied of a good rises when the price of the good rises; and the quantity supplied of a good falls when the price of the good falls

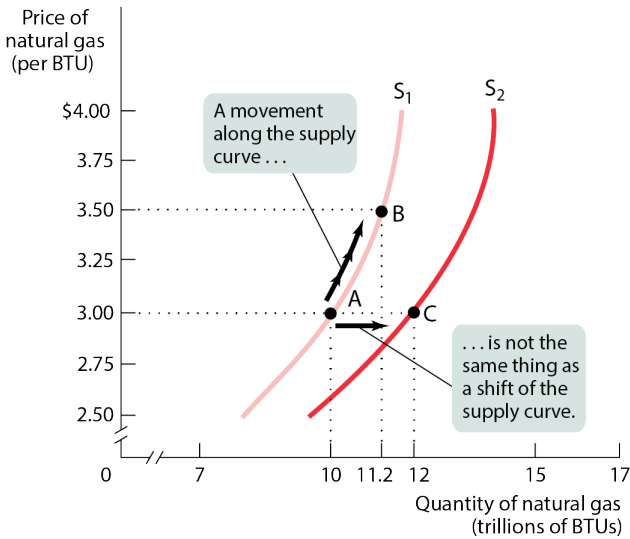
Shifts in the supply curve

A **shift of the supply curve** is a change in the quantity supplied of a good at any given price

- A leftward shift means a drop in supply, a rightward shift means an increase in supply



A **movement along the supply curve** is a change in the quantity supplied of a good arising from a change in the good's price, other things equal



What factors cause the supply curve to shift?

These factors include input prices, the prices of related goods, technology, expectations, and the number of producers

Changes in input prices

An increase in the price of an input of production makes the production of a good more costly, producers are less willing to supply the good at a given price, the supply curve then shifts to the left

- Inputs: labor, raw materials, energy, machinery

Changes in technology

Technology improvement allows producers to raise output without using more inputs, leading to an increase in the supply, i.e., a shift of the supply curve to the right

Examples:

- An innovation in production process, or a reshuffle of management team
- If MacDonald's replaces all workers with robots and automated processes

Changes in expectations

If producers anticipate a higher future price, supply of the good today falls, supply curve today shifts to the left

- But, whether changes in expectations shift supply curves depends on the inventory technology too

Changes in the number of producers

If more producers enter to supply the same good, the *market supply curve* shifts to the right

Recap of demand and supply

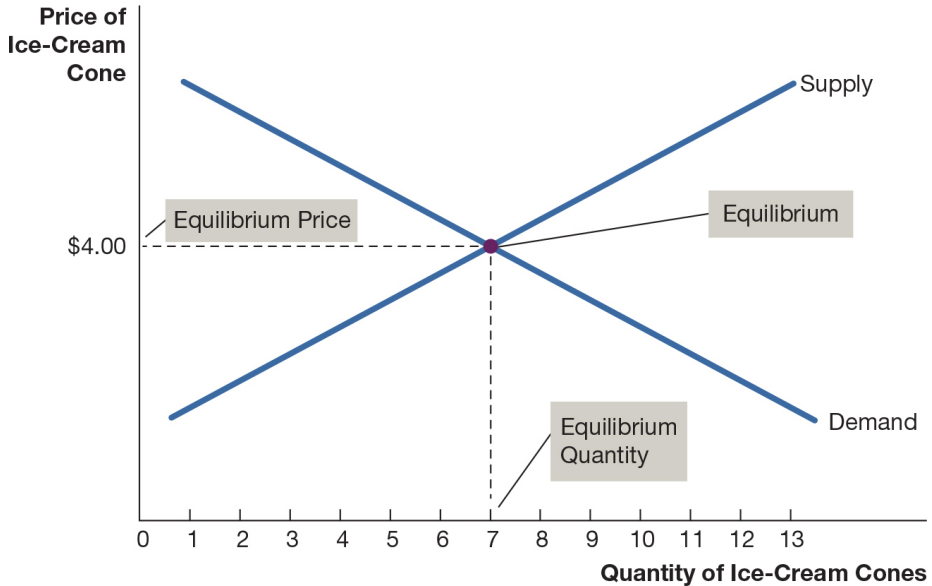
- Difference between quantity demanded and the demand, between quantity supplied and the supply
- Shifts in the demand and supply curves
- Movement along the demand and supply curves is caused by a price change alone

Market Equilibrium

Equilibrium in a perfectly competitive market

Market equilibrium is a situation in which the price has reached the level where quantity supplied equals quantity demanded

- When two quantities equal, we say that **market clears**
- Equilibrium is a point where the demand curve and supply curve intersect.
- The price when quantities demanded and supplied equal is the **equilibrium price** or **market-clearing price**
- The **equilibrium quantity** is the quantity bought and sold at the equilibrium price



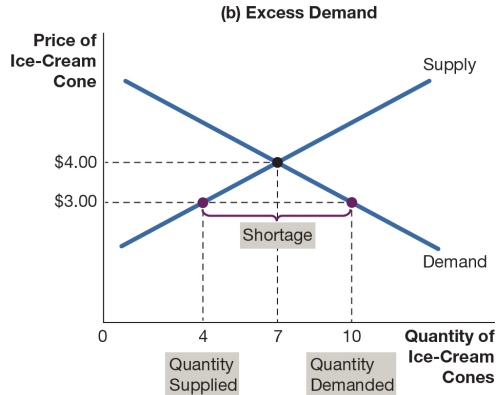
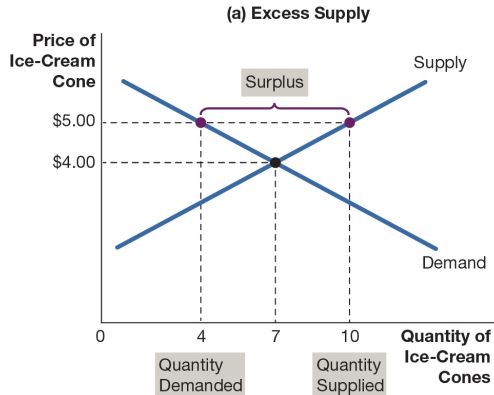
Question:

How can one say that market clears when some buyers do not get what they want at equilibrium price?

Two methods of finding the market equilibrium

1. use diagrams. 2. use algebra

Diagram: an equilibrium can be found using graphs. Market tends to move toward equilibrium



Algebra of market equilibrium

Numerical example. In the market for whatever you fill in,

- the market demand is $P = 22 - 0.5Q^D$
- the market supply is $P = 7 + Q^S$

Three steps of solving for equilibrium:

- Impose equilibrium condition, $Q^D \equiv Q^S = Q$
- Apply the condition to demand and supply, $P = 22 - 0.5Q$ and $P = 7 + Q$
- Solve for price and equilibrium quantity:
- $Q^* = 10$ and $P^* = 17$

Digression. Why no real-life example?

- Data on the demand and supply of a good are mostly not publicly available
- Most goods are not homogeneous goods and their markets are not perfectly competitive
- Data are complex, i.e., they are "contaminated" by many factors that we may not know of
- In fact, empirical analysis of demand and supply is a topic at graduate level

Practice Problem 1

Demand is given by $P = 10 - Q^D$, supply is given by $P = 2 + 0.5Q^S$. Solve for equilibrium. (Answer)

Equilibrium Analysis

Changes in demand and supply

Shifts in the demand curve or supply curve change market equilibrium

The conduct of studying how equilibrium changes is called **equilibrium analysis**

Three steps of equilibrium analysis:

1. Examine whether the event (say a change in income) shifts the demand or the supply or both
2. Decide the direction of curve shifts
3. Use the demand-supply diagram to see how the shift changes equilibrium price and equilibrium quantity

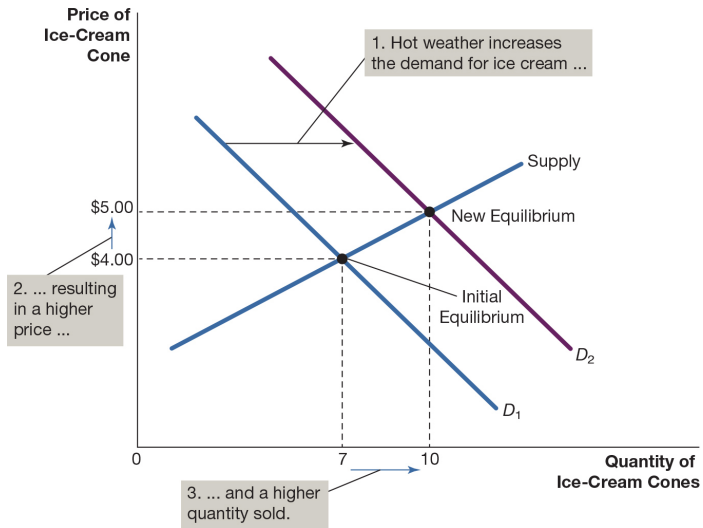
Analysis of equilibrium

For each case of shifts in demand/supply curves,

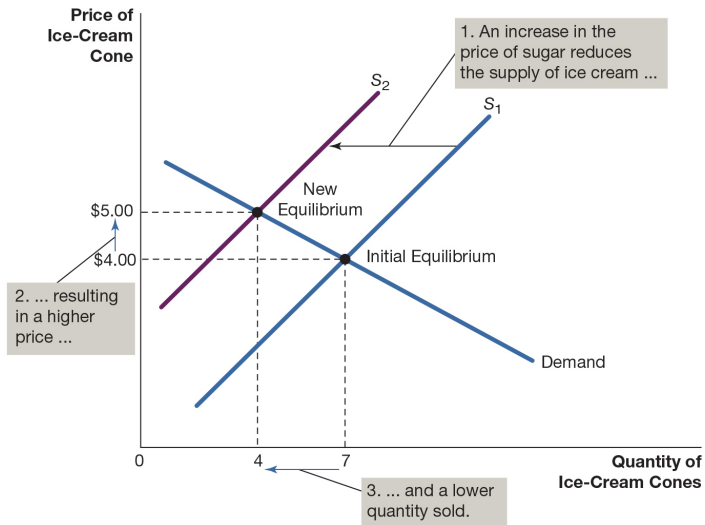
- how does equilibrium price change?
- how does equilibrium quantity change?
- does the change of price or quantity depend by how much demand/supply changes?

Effects of a change in demand

Effects of an increase in demand

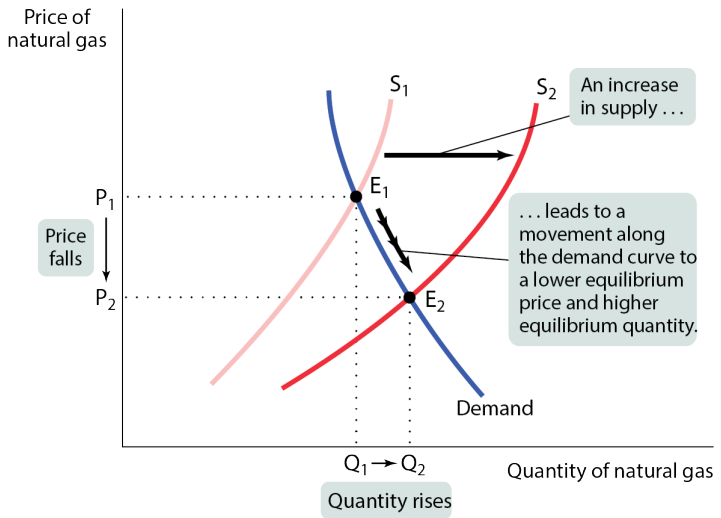


Effects of a decline in Supply

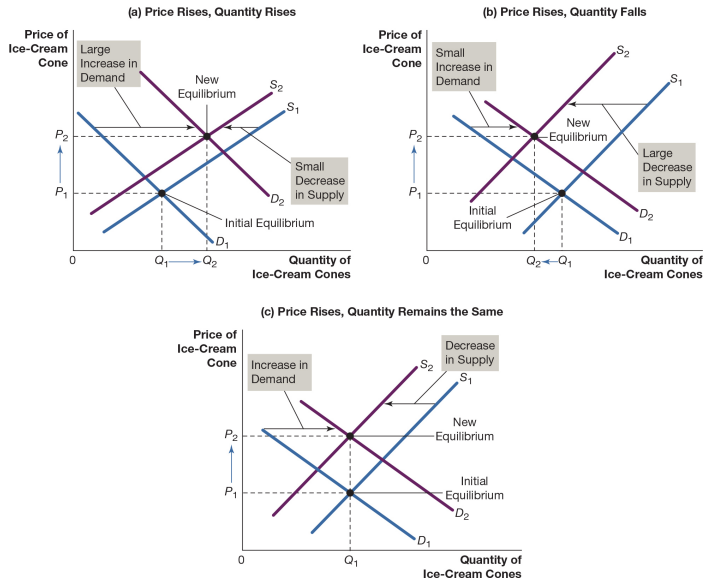


Effects of changes in supply

Effects of an increase in supply



Effects of simultaneous changes in demand and supply



The previous slide shows: increase in demand and fall in supply

Other cases:

- Increase in both demand and supply
- Decrease in both demand and supply
- Decrease in demand, increase in supply

Practice Problem 2:

- For each case of changes in demand and supply, conduct equilibrium analysis

Practice Problem 3

Suppose data show that, when the beef price rises, the demand for broccoli increases

1. Are beef and broccoli substitutes or complements or neither?
2. Following a drop in the beef price, will the price of broccoli rise, fall, or remain unchanged?

(Answer)

Concluding remarks

Concluding remarks

- Analysis of demand and supply in perfectly competitive market
 - Buyers and sellers are price takers
 - Abstract all details of how a market operates
- Regardless of simplification, the equilibrium analysis provides a powerful tool, will be used later
- Market is one way to organize economic activities, price is jointly determined by demand and supply

Practice Problem 1

In equilibrium, it must be $Q^D = Q^S = Q$. Using this condition and from equations for demand and supply, we have $P = 10 - Q$ and $P = 2 + 0.5Q$. Two equations imply that $10 - Q = 2 + 0.5Q$. Add Q to both sides of the previous equation, we get $10 = 2 + 1.5Q$. Subtract 2 from both sides of the equation, we have $8 = 1.5Q$

Equilibrium quantity is $Q^* = \frac{8}{1.5} = 5.33$. Equilibrium price is then $P^* = 10 - 5.33 = 4.67$

(Back to [question](#))

Practice Problem 3

Suppose data show that, when the beef price rises, the demand for broccoli increases

1. Substitutes
2. Broccoli price will fall

(Back to [question](#))